S.NO	Item	Instrument ID	Group Marking	Operating Voltage	KW	Amps	MPCB/MCCB Rating	MPCB/MCCB	Total Qty	Total KW	Req Inverter	Req DOL	Req Soft Starter	Req DI	Req DO	Req AI 4-20mA	Req AI 0-10V/4-20mA	Req AO 4-20mA
Panel Specification WNS2503																		
BIO/UF Panel Requirements																		
Biological Section Pumps in BIO/UF Panel																		
1	Primary Screening	S101	S101	3PH 400V 50HZ	0.37	0.70	0.9 To 1.25A	MPCB-3RV	1	0.37		1		2	1			
2	Oil Skimmer	S102	S102	3PH 400V 50HZ	1.5	2.85	2.8 - 4A	MPCB-3RV	1	1.5		1						
3	Equlaisaton Blower	K101A/B	K101A/B	3PH 400V 50HZ	15	28.52	28 - 40A	MPCB-3RV	2	30	2							
4	Remote Switch of EQ Blower	RS_K101A/B	RS_K101A/B	24V DC					2	0				2	0			
5	Biological Feeding Pump(cooling Tower Feed Pumps)	P101A/B	P101A&P101B	3PH 400V 50HZ	4	7.60	7-10A	MPCB-3RV	2	8	2							
6	Cooling Tower	E101	E101	3PH 400V 50HZ	7.5	14.26	11 - 16A	MPCB-3RV	1	7.5			1	6	4			
7	Biological Oxidation Blower	K102A/B	K102A&K102B	3PH 400V 50HZ	37	70.34	100A	MPCB-3RV	2	74	2							
8	Remote Switch for Oxy Blower	K102A/B	RS_K102A&RS_K102B_	24V DC					2	0				2	0			
9	Acid Dosing Pump for Biological pH correction	P102A/B	P102A &P102B	1PH 230V 50HZ	0.75	1.43	1.4 To 2A	MPCB-3RV	2	1.5		2		4	2			
10	Bio Sludge Recurculation Pump	P103A/B	P103A&P103B	3PH 400V 50HZ	2.2	4.18	3.5 - 5A	MPCB-3RV	3	6.6	2							
11	Underground Drainage Pump	P104A/B	P104A & P104B	3PH 400V 50HZ	1.9	3.61	2.8 - 4A	MPCB-3RV	2	3.8		2		4	2			
Biological Section Instruments in BIO/UF Panel										0								
1	Emergency Switch_Panel	ES_MP		24V DC					1	0				1				
2	Emergency Switch Remote_BIO/UF	ES_R1	BIO_RES_100	24V DC					1	0				1				
3	Reset Switch _Panel	RS_01		24V DC					1	0				1				
4	Tower Lamp With Buzzer	TL_MP		24VDC					1	0					2			
5	Thermostat For Panel	TS_P	TS_BIO_VFD	24V DC					1	0				1				
6	Primary Screen Level Switch	LS100	LS101	24V DC					1	0				1				
7	Clarifier Permeate Tank Level	LS102	LS102	24V DC					6	0				6				
8	Sludge Reciculation Tank Level	LS101	LS101	24V DC					3	0				3				
9	Underground Drainage Tank Level	LS103	LS103	24V DC					3	0				3				
10	Level Transmitter On Biological Feeding Tank	LT101	LT101	4-20mA					1	0						1		
11	Level Transmitter on Acid Tank	LT102	LT102	4-20mA					1	0						1		
12	Temerature Transmitter Cooling Tower Outlet	TT101	TT101	4-20mA					1	0						1		
13	pH Transmitter Bio Feed	AIT101	AIT101_P &AIT101_C	230V,4-20mA					1	0						2		
14	DO Transmitter_Tank A&B	AIT102A/B	AIT102A _P&AIT102A_C &AIT102B_P and AIT102B_C	230V,4-20mA					2	0						4		
15	Flow Meter Bio Feed	FIT101	FIT101_P & FIT101_C	230V,4-20mA					1	0				1		1		
16	Electro Valve for EQ Air Line	EV101	EV101	230V AC					1	0					1			
17	Electro Valve for Bio-Air Line	EV102	EV102	230V AC					1	0					1			
Sand Fitler and UF Pumps and Blower										0								
1	Drum Screen	S102	S102	3PH 400V 50HZ	1.1	2.09	2.2 - 3.2A	MPCB-3RV	1	1.1		1		2	1			
2	Sand Filter Feed and Backwash Pumps	P201A/B/C	P201A &P201B&P201C	3PH 400V 50HZ	7.5	14.26	11 - 16A	MPCB-3RV	3	22.5	3							
3	Remote Switch for Sand Filter Feed and Backwash Pump Remote Switch	RS_P201A/B	RS_P201A &RS_P201B&RS_P201C	24V DC		0.00			3	0				3				
4	Sand Filter and UF Backwash Blower	K201A/B	K201A &K201B	3PH 400V 50HZ	7.5	14.26	11 - 16A	MPCB-3RV	2	15	2							
5	Remote Switch For MBR Blowers	RS_K201A/B	RS_K201A and RS_201B	24VDC		0.00			2	0				2				
6	UF Feed and CIP Pumps	P202A/B/C	P202A&P202B&P202C	3PH 400V 50HZ	7.5	14.26	11- 16A	MPCB-3RV	3	22.5	3							
7	Softener Feed Pump	P203A/B	P203A&P203B	3PH 400V 50HZ	7.5	14.26	11- 16A	MPCB-3RV	2	15	2							
8	Softener Regenation Pump	P204A/B	P204A & P204B	3PH 400V 50HZ	3	5.70	4.5 - 6.3A	MPCB-3RV	2	6		2		4	2			
9	UF Backwash Pump	P205A/B	P205A & P205B	3PH 400V 50HZ	11	20.91	17 - 22A	MPCB-3RV	2	22	2							
10	UF Reject Transfer Pump	P206A/B	P206A & P206B	3PH 400V 50HZ	1.5	2.85	2.8 -4A	MPCB-3RV	2	3	2							
11	De-Gasser Blower	K202	K202	3PH 400V 50HZ	3.7	7.03	5.5 To 8A	MPCB-3RV	1	3.7		1		2	1			
12	Citric Dosing Tank Mixer	MX401	MX401 & MX401B	3PH 400V 50HZ	1.5	2.85	2.8 - 4A	MPCB-3RV	1	1.5		1		2	1			
13	Sodium Hydroxide Dosing Tank Mixer	MX402	MX402 & MX402B	3PH 400V 50HZ	1.5	2.85	2.8 - 4A	MPCB-3RV	1	1.5		1		2	1			
14	HCL Dosing for Softner Regeneration	P405A/B	P405A & P405B	3PH 400V 50HZ	1.5	2.85	2.8 - 4A	MPCB-3RV	2	3		2		4	2			
15	Hypo Dosing Pump for Sand Filter Feed Tank	P407A/B	P407A & P407B	1PH 230V 50HZ	0.25	1.36	1.4 - 2A	MPCB-3RV	2	0.5		2		8	6			2
16	Sodium Hydroxide Dosing For UF CEB	P411A/B	P411A & P411B	1PH 230V 50HZ	0.25	1.36	1.4 - 2A	MPCB-3RV	2	0.5		2		8	6			2
17	Caustic Dosing Pump for RO Feed pH Correction	P412A/B	P412A & P412B	1PH 230V 50HZ	0.25	1.36	1.4 - 2A	MPCB-3RV	2	0.5		2		8	6			2
18	Caustic Dosing Pump for Softener Regeneration	P413A/B	P413A & P413B	3PH 400V 50HZ	0.75	1.43	1.4 - 2A	MPCB-3RV	2	1.5		2		4	2			
Instruments Sand and UF Filter Section in Main Panel										0								
1	Sand Filter Permeate Tank Level	LS201	LS201	24V DC					6	0				6				
2	UF Permeate Tank Level Switch	LS202	LS202	24V DC					6	0				6				
3	RO Feed Tank Level Switch	LS203	LS203	24V DC					6	0				6				
4	UF CIP Tank Safety Level Switch	LS204	LS204	24V DC					1	0				1				
5	UF Reject Tank Level Switch	LS205	LS204	24V DC					5	0				5				
6	Thermostat Signal from CIP Heater	TS_201	TS201	24V Dc					1	0				1				
7		TT201	TT201	24V DC/4-20mA					1	0						1		
8	Sand Filter Feed Flow Meter	FIT201	FIT201_P & FIT201C	230V,4-20mA					1	0				1		1		
9	UF Feed Flow Meter	FIT202	FIT202_P & FIT202C	230V,4-20mA					1	0				1		1		
10	UF Permeate Flow Meter	FIT203	FIT203_P & FIT203C	230V,4-20mA					1	0				1		1		
11	Softener Feed Flow Meter	FIT204	FIT204_P & FIT204C	230V,4-20mA					1	0				1		1		
12	Soft Regeneration Flow Meter	FIT205	FIT205_P & FIT205C	230V,4-20mA					1	0				1		1		
13	UF CIP Flow Meter	FIT206	FIT206_P & FIT205C	230V,4-20mA					1	0				1		1		
14	UF Reject Transfer Flow Meter	FIT207	FIT207_P & FIT207C	230V,4-20mA					1	0				1		1		
15	Sand Filter Feed Pressure Transmiitter	PIT201	PIT201	4-20mA					1	0						1		
16	Sand Filter Outlet Pressure Transmitter	PIT202	PIT202	4-20mA					1	0						1		
17	UF Backfilter Inlet Pressure Transmitter	PIT203	PIT203	4-20mA					1	0						1		
18	UF Backfilter Outlet Pressure Transmitter	PIT204	PIT204	4-20mA					1	0						1		
19	UF Permeate Pressure Transmitter	PIT205	PIT205	4-20mA					1	0						1		
20	Softener Feed Pressure Transmitter	PIT206	PIT206	4-20mA					1	0						1		
21	Softener Regeneration Pressure Transmitter	PIT207	PIT207	4-20mA					1	0						1		
22	Softener Outlet Pressure Transmitter	PIT208	PIT208	4-20mA					1	0						1		
23	UF CIP pH Transmitter	AIT202	AIT202_P & AIT202_C	230V,4-20mA					1	0						2		
24	Pneumatic Panel UF_CP01	CP01		24V DC/0-10V					1	0				1			1	
25	Pneumatic Valve Feed Back	PV201 To PV250							50	0				100				
26	Citric Dosing Tank Level Transmitter	LT402		4-20mA					1	0						1		
27	Hypo Dosing Tank Level Switch	LT404		4-20mA					1	0				1		1		
28	HCL Dosing Tank Level Transmitter	LT403		4-20mA					1	0						1		
29	Sodium Hydroxide Dosing Level Transmitter	LT405		4-20mA					1	0						1		

UF Reject Chemical Treatment										0									
1	UF Reject Decorbanation Tank _1 Mixer	MX701	MX701	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
2	UF Reject Decorbanation Tank _2 Mixer	MX702	MX702	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
3	Coagulant Mixer	MX703	MX703	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
4	Sodium Carbonate Dosing Mixer	MX704	MX704	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
5	Poly Dosing Tank Mixer for Decorbanation	MX705	MX705	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
6	Sodium Hydroxide Dosing Mixer	MX706	MX706	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
7	ferrichloride Dosing Pump	P419A/B	P701A&P701B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		8	6			2
8	Sodium Carbonate Dosing Pump	P702A/B	P702A&P702B	3PH 400V 50HZ	0.25	0.48	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			
9	Poly Dosing pump for Flocculation	P418A/B	P703A&P703B	3PH 400V 50HZ	0.25	0.48	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			
10	Sodium Hydroxide Dosing Pump	P704A/B	P704A&P704B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		8	6			2
UF Reject Instruments										0								
1	Coagulant Dosing Tank Level Transmitter	LT701		4-20mA					1	0				1				
2	Sodium carbonate dosing Tank Level	LT702		4-20mA					1	0				1				
3	Poly dosing Tank Level Transmitter for Decorbanation	LT703		4-20mA					1	0						1		
4	Electro Valves	EV701 to EV705		230V AC					5	0					5			
	BIO_Sand Filter_UF Panel Total									264.1	22	36	1	259	68	33	1	10
RO PANEL REQUIREMENTS																		
1st RO Section Pumps																		
1	RO Feed Pump	P301A/B	P301A &P301B	3PH 400V 50HZ	11	20.91	20 -25A	MPCB-3RV	2	22			2	10	6			
2	Remote Switch for RO Feed Pump	RS_P301A/B	RS_P310A &RS_P310B			0.00			2	0				2				
3	RO High Pressure Pumps -1st stage	P302A/B	P302A & P302B	3PH 400V 50HZ	30	57.03	45- 63A	MCCB-3VM	2	60	2							
4	Remote Switch for HP Pumps	RS_P302A/B	RS_P302	24V DC		0.00			2	0				2				
5	RO Booster Pumps - 2nd Stage	P303A/B	P303A&P303B	3PH 400V 50HZ	5.5	10.46	9 -12.5A	MPCB-3RV	2	11	2							
6	Remote Switch for RO Booster Pumps	RS_P303A/B	RS_P303	24V DC		0.00			2	0				2				
7	RO Booster Pumps - 3rd Stage	P304A/B	P304A & P304B	3PH 400V 50HZ	3	5.70	4.5 - 6.3A	MPCB-3RV	2	6	2							
8	Remote switch for RO Booster Pumps -3rd Stage	RS_P304A/B	RS_P304	24V Dc		0.00			2	0				2				
9	CIP Make up Pump	P305A/B	P305A & P305B	3PH 400V 50HZ	5.5	10.46	9 - 12.5A	MPCB-3RV	2	11		2		4	2			
10	CIP Make up Pump Remote Switch	RS_P305A/B	RS_P305A & P305B	24V DC		0.00			1	0				2	6			
11	CIP Pumps - RO	P401A/B	P401A &P401B	3PH 400V 50HZ	7.5	14.26	11 - 16A	MPCB-3RV	2	15	2							
12	Remote Switch For CIP Pumps	RS_P401A/B	RS_P401A&RS_401B&RS	24V DC		0.00			2	0				2				
13	RO CIP Heater	HT401	HT401	3PH 400V 50HZ	15	28.52	27 To 32 A	MPCB-3RV	1	15		1		2	1			
14	Caustic Dosing Pump for RO Feed pH Correction	P412A/B	P412A & P412B	1PH 230V 50HZ	0.25	1.36	1.1 To 1.6A	MPCB-3RV	2	0.5		2		8	6			2
15	RO_1 Antiscalent Dosing Pumps	P414A/B	P402A&P402B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			2
16	RO_1 Antiscalent Dosing Pumps Control	P414A/B	P402A_C & P402B_C	24V DC		0.00			2	0				4	2			
17	RO_1 SMBS Dosing Pumps	P416A/B	P403A & P403B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			2
18	RO-1 SMBS Dosing Pumps Control	P416A/B	P403A_C &P403B&C	24V DC		0.00			2	0				4	2			
RO Instruments										0								
1	Emergency Switch on RO Panel	ES_MP		24V DC					1	0				1				
2	Emergency Button Remote	ES_2	RES_200	24V DC					1	0				1				
3	Reset On Panel	RS_02		24 V DC					1	0				1				
4	RO Feed Conductivity Transmitter	AIT301	AIT303_P&AIT303_C	230V,4-20mA					1	0						2		
5	RO Feed Redox Transmitter	AIT302	AIT302_P&AIT302_C	230V,4-20mA					1	0						1		
6	RO Feed pH correction	AIT303	AIT301_P&AIT301_C	230V,4-20mA					1							2		
7	RO Total Permeate Conductivity Transmitter	AIT304	AIT304_P&AIT304_C	230V,4-20mA					1	0						1		
8	RO Feed Flow Meter	FIT301	FIT301_P&FIT301_C	230V,4-20mA					1	0				1		1		
9	RO Total Permeate Flow Meter	FIT302	FIT302_P&FIT302_C	230V,4-20mA					1	0				1		1		
10	RO Final Reject Flow Meter	FIT303	FIT303_P&FIT303_C	230V,4-20mA					1	0				1		1		
11	RO 1st Stage Permeate Flow Meter	FIT304	FIT304_P&FIT304_C	230V,4-20mA					1	0				1		1		
12	RO 2nd Stage Permeate Flow Meter	FIT305	FIT305_P&FIT305_C	230V,4-20mA					1	0				1		1		
13	RO 3rd Stage Permeate Flow Meter	FIT306	FIT306_P&FIT306_C	230V,4-20mA					1	0				1		1		
14	RO 4th Stage Permeate Flow meter	FIT307	FIT307_P&FIT373_C	230V,4-20mA					1	0				1		1		
15	RO CIP Flow Meter	FIT401	FIT308_P&FIT308_C	230V,4-20mA					1	0				1		1		
16	RO Feed pH correction	AIT401	AIT401_P&AIT401_C	230V,4-20mA					1							2		
17	RO Feed Catriage Filter Inlet Pressure	PIT301	PIT301	4-20mA					1	0						1		
18	RO Feed Catriage Filter Outlet Pressure	PIT302	PIT302	4-20mA					1	0						1		
19	Pressure Transmitter - RO 1st Stage Inlet	PIT303	PIT303	4-20mA					1	0						1		
20	Pressure Transmitter - RO 1st Stage Outlet	PIT304	PIT304	4-20mA					1	0						1		
21	Pressure Transmitter RO 2nd Stage Inlet	PIT305	PIT305	4-20mA					1	0						1		
22	Pressure Transmitter RO 2nd Stage outlet	PIT306	PIT306	4-20mA					1	0						1		
23	Pressure Transmitter RO 3rd Stage Inlet	PIT307	PIT307	4-20mA					1	0						1		
24	Pressure Transmitter RO 3rd Stage Outlet	PIT308	PIT308	4-20mA					1	0						1		
25	Pressure Transmitter RO 4th Stage Inlet	PIT309	PIT309	4-20mA					1	0						1		
26	Pressure Transmitter RO 4th Stage Outlet	PIT310	PIT310	4-20mA					1	0						1		
27	RO Total Permeate Pressure Transmitter	PIT311	PIT311	4-20mA					1	0						1		
28	RO Permeate Tank Level Swtich	LS301	LS301	24V DC					6	0				6				
29	RO Reject Tank Level Switch	LS302	LS302	24V DC					6	0				6				
30	RO CIP-1 Level Trasmitter	LT401	LT401	4-20mA					1	0				1		1		
31	RO CIP Temp Transmitter	TT401	TT401	4-20mA					1	0						1		
32	RO CIP Thermostat	TS401	TS401	24VDC					1	0				1				
33	SMBS Dosing Tank Level Switch	LS405	LS401	24V DC					1	0				1				
34	Antiscalent Dosing Tank Level Switch	LS404	LS402	24V DC					1	0				1				
35	Pneumatic Valve Feed Back Limit Switch	PV301to PV360	PV301 & PV302 &PV303....PV310	24V DC					65	0				130				
36	RO Pneumatic Panel	CP02	RO_CP02	24V DC/0-10V					1	0				1			1	4
RO Reject Chemical Treatment										0								
1	RO Reject water Transfer Pump	P306A/B	P306A & P306B	3PH 400V 50HZ	0.75	1.43	1.4 To 2A	MPCB-3RV	2	1.5	2							
2	RO Reject Decorbanation Tank _1 Mixer	MX501	MX501	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
3	RO Reject Decorbanation Tank _2 Mixer	MX502	Mx502	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
4	RO Reject Decorbanation Tank _3 Mixer	MX503	MX503	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
5	Coagulant Mixer	MX504	MX504	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
6	Sodium Carbonate Dosing Mixer	MX505	MX505	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
7	Poly Dosing Tank Mixer for Decorbanation	MX506	MX506	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
8	Sodium Hydroxide Dosing Mixer	MX507	MX505	3PH 400V 50HZ	1.5	2.71	1.8 To 2.5 A	MPCB-3RV	1	1.5		1		2	1			
9	Coagulant Dosing Pump	P501A/B	P501A & P501B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		8	6			2

10	ferrichloride Dosing Pump	P502A/B	P502A & P502B	3PH 400V 50HZ	0.25	0.48	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			
11	Poly Dosing pump for Flocculation	P503A/B	P503A & P503B	3PH 400V 50HZ	0.25	0.48	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			
12	CRO PH Correction Pump	P504A/B	P504A & P504B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			2
13	CRO PH Correction Control	P504A/B	P405A_C & P405B_C	24V DC		0.00			2	0				4	2			
14	MG source Dosing Pump	P505A/B	P505A & P505B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		8	6			2
15	Chemical sludge Transfer Pump to Screw Press	P506A/B	P506A & P506B	3PH 400V 50HZ	1.5	2.85	2.8 To 4A	MPCB-3RV	2	3		2		4	2			
16	Hypo Dosing Pump for Sand Filter Feed Tank	P408A/B	P408A & P408B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		8	6			2
17	Extra DOL	P507A/B/C/D	P507A & P507B & P507C& P507D	3PH 400V 50HZ	4	7.22	7 To 10A	MPCB-3RV	4	16		4		8	4			
RO Reject Instruments										0								
1	CRO Sand Filter Feed Tank Level Switch	LS501		24V DC					6	0				6				
2	Chemical Sludge Tank Level Switch	LS502		24V DC					3					3				
3	Coagulant Dosing Tank Level Transmitter	LT501		4-20mA					1	0				1				
4	Sodium carbonate dosing Tank Level	LT502		4-20mA					1	0				1				
5	Poly dosing Tank Level Transmitter for Decorbanation	LT503		4-20mA					1	0						1		
6	Sulphuric Acid Dosing Tank Level Transmitter	LT504		4-20mA					1	0				1				
7	Sodium Hydroxide Level Transmitter	LT505		4-20mA					1	0				1				
8	RO Reject Decorbanation Tank pH meter	AIT501	AIT501_P &AIT501_C	230V,4-20mA					1	0						1		
9	CRO Feed pH meter	AIT502	AIT502_P & AIT502_C	230V,4-20mA					1	0						1		
10	RO Final Reject Flow Meter	FIT309	FIT309_P&FIT309_C	230V,4-20mA					1	0				1		1		
11	Electro Valves	EV501 to EV505		230V AC					5						5			
CRO Section Pumps										0								
1	CRO Feed Pump	P601A/B	P601A & P601B	3PH 400V 50HZ	2.2	4.18	4.5 - 6.3A	MPCB-3RV	2	4.4		2		4	2			
2	Remote Switch for CRO Feed Pump	RS_601A/B	RS_P601A &RS_P601B			0.00			2	0				2				
3	CRO High Pressure Pumps -1st stage_Pump1	P602A/B	P602A & P602B	3PH 400V 50HZ	7.5	14.26	11 - 16A	MPCB-3RV	2	15	2							
4	Remote Switch for HP Pumps	RS_602A/B	RS_P602	24V DC		0.00			2	0				2				
5	CRO High Pressure Pumps -1st stage_Pump2	P603A/B	P603A & P603B	3PH 400V 50HZ	7.5	14.26	11- 16A	MPCB-3RV	2	15	2							
6	Remote Switch for RO Booster Pumps	RS_603A/B	RS_P603	24V DC		0.00			2	0				2				
7	CRO Reject Recycle Pump	P604A/B	P604A & P604B	3PH 400V 50HZ	4	7.60	7 to 10A	MPCB-3RV	2	8	2							
8	Remote Switch for RO Booster Pumps	RS_603A/B	RS_P603	24V DC		0.00			2	0				2				
9	RO_1 Antiscalent Dosing Pumps	P415A/B	P415A & P415B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			2
10	RO_1 Antiscalent Dosing Pumps Control	P415A/B	P402A_C & P402B_C	24V DC		0.00			2	0				4	2			
11	RO_1 SMBS Dosing Pumps	P417A/B	P417A & P417B	1PH 230V 50HZ	0.25	1.36	1.1A To 1.6A	MPCB-3RV	2	0.5		2		4	2			2
12	RO-1 SMBS Dosing Pumps Control	P417A/B	P403A_C &P403B&C	24V DC		0.00			2	0				4	2			
CRO Instruments										0								
1	CRO Sand Filter Backwash Flow Meter	FIT308	FIT309_P&FIT309_C	230V,4-20mA					1	0				1		1		
2	CRO Feed Conductivity Transmitter	AIT601	AIT602_P&AIT602_C	230V,4-20mA					1	0						2		
3	CRO Total Permeate Conductivity Transmitter	AIT602	AIT603_P&AIT603_C	230V,4-20mA					1	0						1		
4	CRO Feed Flow Meter	FIT601	FIT601_P&FIT601_C	230V,4-20mA					1	0				1		1		
5	CRO Final Reject Flow Meter	FIT602	FIT602_P&FIT602_C	230V,4-20mA					1	0				1		1		
6	CRO Total Permeate Flow Meter	FIT603	FIT603_P&FIT603_C	230V,4-20mA					1	0				1		1		
7	CRO Recirculation Flow Meter	FIT604	FIT604_P & FIT604_C	230V,4-20mA					1	0				1		1		
8	CRO Feed Catriage Filter Inlet Pressure	PIT601	PIT601	4-20mA					1	0						1		
9	CRO Feed Catriage Filter Outlet Pressure	PIT506	PIT506	4-20mA					1	0						1		
10	Pressure Transmitter - CRO 1TH Stage Inlet	PIT603	PIT603	4-20mA					1	0						1		
11	Pressure Transmitter - CRO 1st Stage Outlet	PIT604	PIT604	4-20mA					1	0						1		
12	Pressure Transmitter CRO 6th Stage Inlet	PIT605	PIT605	4-20mA					1	0						1		
13	Pressure Transmitter CRO 6th Stage outlet	PIT606	PIT606	4-20mA					1	0						1		
14	CRO and Reject Section Pneucmatic Panel	CP03	CP03						1	0				1		1		
15	Final RO Reject Tank Level	LS601	LS601						6	0				6				
RO Panel Total										**218.9**	**16**	**40**	**2**	**330**	**83**	**48**	**1**	**22**